

# MD. AKASH MIAH

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## PROFESSIONAL SUMMARY

AI/ML Engineer with hands-on experience in machine learning model development, deployment, and optimization. Skilled in building end-to-end AI systems using Python, PyTorch, and TensorFlow. Strong background in computer vision, LLM-based systems, and data-driven problem solving, with experience translating real-world requirements into production-ready AI solutions.

## TECHNICAL SKILLS

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|-----------------------------|-------------------------------------------------------------------------------------------------|
| Programming & ML Frameworks | Python, Machine Learning, Deep Learning, Model Training & Evaluation, Optimization              |
| AI Techniques               | PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, OpenCV                                        |
| Data Science                | Computer Vision, NLP, Transformers, Fine-tuning, Prompt Engineering                             |
| Tools                       | EDA, Statistical Analysis, Regression, Model Validation                                         |
| LLMs & GenAI                | Docker, Git, SQL, Google Colab, Python Socket Programming, FastAPI                              |
|                             | LLM Inference, LangChain, N8n, RAG, Qdrant, PGvector, Pinecone, Chroma, OpenAI/Gemini/Groq APIs |

## PROFESSIONAL EXPERIENCE

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| <b>AI Engineer</b><br><i>Betopia LTD</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Jul 2025 – Present<br><i>Dhaka, Bangladesh</i> |
| <ul style="list-style-type: none"><li>Contributed to transforming Betopia into an <b>AI-first platform</b> through LLM systems and intelligent tooling</li><li>Built and deployed an internal <b>LLM provider platform</b>, enabling unified API access to multiple models</li><li>Developed an internal <b>coding agent</b>, leveraging proprietary LLMs to automate engineering workflows</li><li>Fine-tuned <b>transformer-based models</b> for domain-specific tasks on custom datasets</li><li>Optimized <b>LLM inference pipelines</b>, reducing latency and improving response consistency</li><li>Designed <b>scalable backend systems</b> for AI services, supporting real-time inference and integrations</li></ul> |                                                |

## EDUCATION

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|---------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| <b>B.Sc. in Computer Science and Engineering</b><br><i>North South University</i>                                         | 2025<br><i>Dhaka, Bangladesh</i> |
| <b>Professional Development:</b> Short Course on Data Science — Prof. Jennifer Widom, Stanford University <i>Feb 2024</i> |                                  |

## KEY PROJECTS

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                               |
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| <b>Betopia AI Ecosystem</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <i>Go, Python, TypeScript, vLLM, PostgreSQL, Redis, React</i> |
| <ul style="list-style-type: none"><li>Architected and led development of a <b>three-product AI ecosystem</b> as founding AI team member; currently in internal beta with planned public launch</li><li><b>Betopia AI Chat</b> — Conversational AI assistant with multi-turn memory, web search, file analysis, and Bengali language support; deployed on self-hosted vLLM infrastructure</li><li><b>Betopia Coder</b> — AI coding agent with multi-agent pipeline, Go HTTP backend, and LLM abstraction layer supporting multiple providers</li><li><b>Betopia AI API Platform</b> — Developer-facing inference API with JWT/API key auth, PostgreSQL, Redis rate limiting, Prometheus metrics, RBAC, and cloud provider fallback routing</li></ul> |                                                               |
| <b>AI-Based Grocery Checkout System</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <i>Python, OpenCV, YOLOv8, RT-DETR</i>                        |
| <ul style="list-style-type: none"><li>Built an AI-powered checkout system that detects and classifies <b>10 grocery product classes</b> from a single image</li><li>Trained YOLOv8 on a custom dataset, achieving <b>0.995 mAP@50</b> on the validation set</li><li>Optimized inference pipeline to <b>696 ms/frame</b>, enabling near real-time performance</li><li>Designed system to simulate barcode scanning by mapping detections to <b>GTIN-based product IDs</b></li></ul>                                                                                                                                                                                                                                                                  |                                                               |
| <b>Autonomous Chemical Spraying Robot</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <i>YOLOv8, IoT, Edge AI</i>                                   |
| <ul style="list-style-type: none"><li>Developed an autonomous agricultural robot combining <b>computer vision, IoT, and robotics</b></li><li>Deployed YOLOv8 on edge hardware for real-time crop and weed classification</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                               |

- Integrated perception outputs with robotic actuation to enable **targeted chemical spraying**
- Improved spraying efficiency by minimizing unnecessary chemical usage